Performed by: Vipul Jadhav

Assignment 5

Task: Clone a GitHub repository into the Jenkins master and use Maven to build the project and generate a WAR file from the pom.xml. Upload each build to an S3 bucket with a timestamped name (e.g., LoginWebApp_DATE_TIME.war) so that every build is stored. Finally, download the latest WAR file from S3 to the Tomcat webapps directory and deploy it as LoginWebApp.war.

Steps:

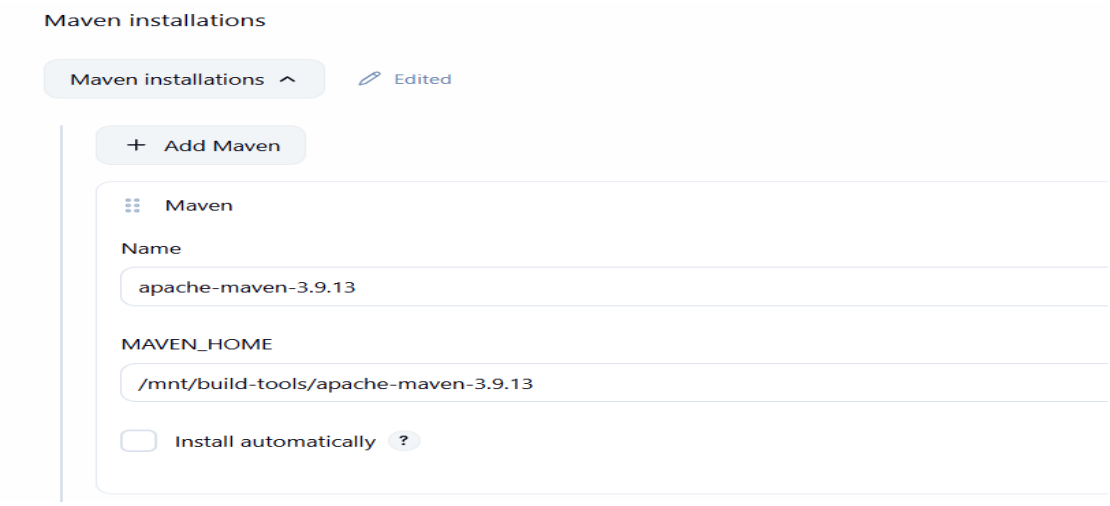
1. Install maven in Jenkins master.

```
[root@ip-172-31-33-28 apache-maven-3.9.13]# pwd
/mnt/build-tools/apache-maven-3.9.13
```

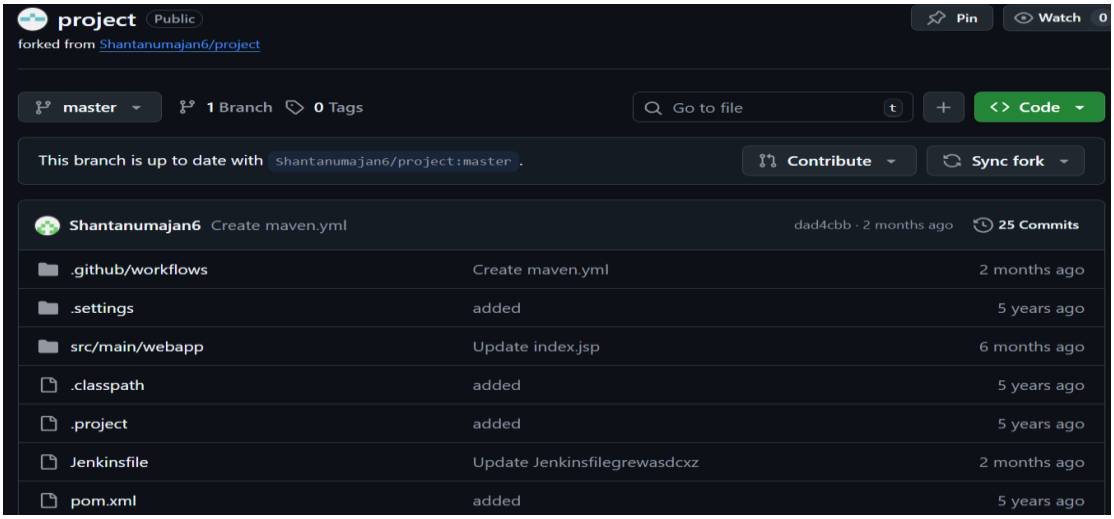
2. Add maven to environment variables

```
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
        . ~/.bashrc
fi

# User specific environment and startup programs
#
#
export MAVEN_HOME=/mnt/build-tools/apache-maven-3.9.13
export PATH=$MAVEN_HOME/bin:$PATH
```

3. Uncheck install maven automatically checkbox and add name of maven (added apache-maven-3.9.13) and add the location of maven home.

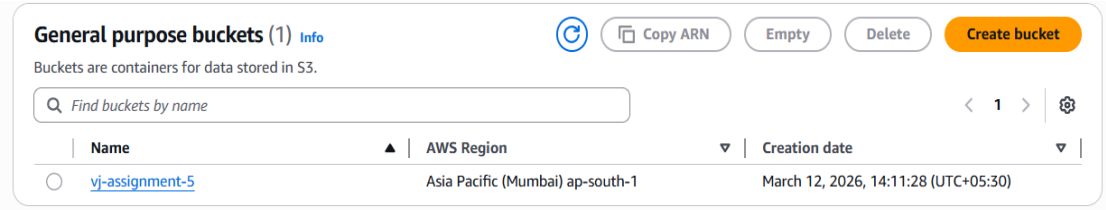


4. Github repository that we are going to use.

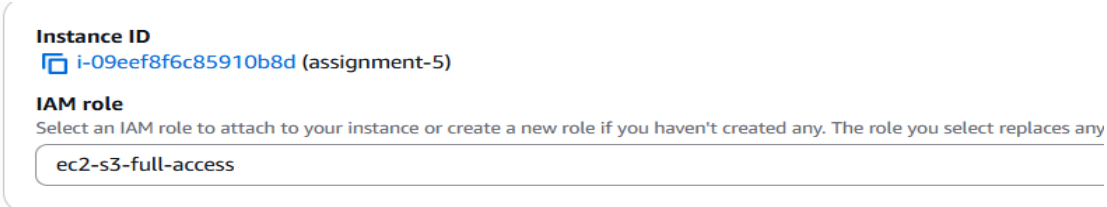


5. Install git on Jenkins home

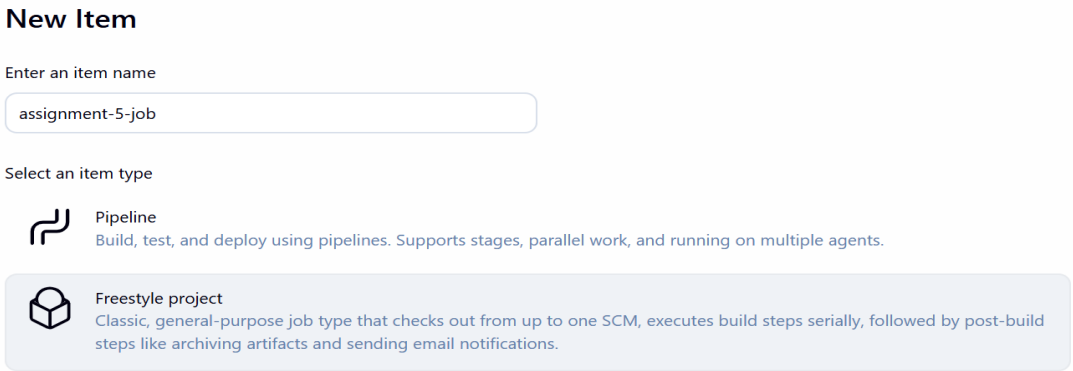
6. Create new s3 bucket



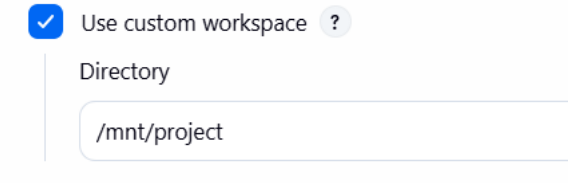
7. attach ec2 to s3 full access iam role to the Jenkins master so that we can access the instance using cli



8. Create a freestyle project job on Jenkins with the name assignment-5-job

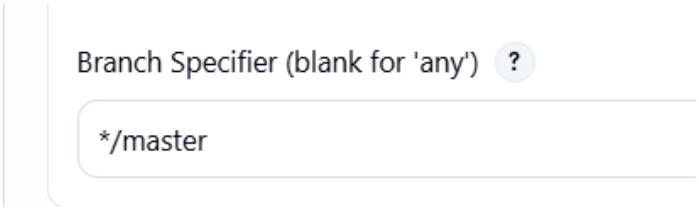


9. Added custom workspace

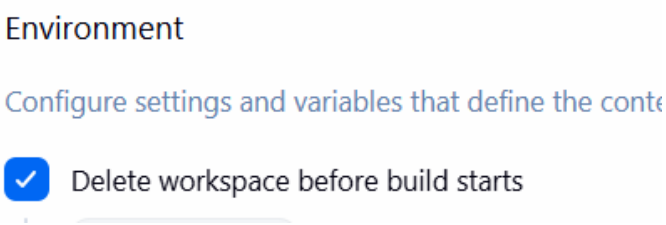


10. Put github repository URL in github URL section of Source Code Management

11. Select master branch in Branch Specifier



12. Check on 'Delete workspace before build starts' option. So, every time when we start new build the workspace will be cleared



13. Now in build step wrote first shell script to clean the maven repository

14. In build step selected Invoke top-level Maven targets configured Maven version and specified the goal in which mentioned two maven tasks (clean and install)

Invoke top-level Maven targets ?

Maven Version

apache-maven-3.9.13

Goals

```
clean
install
```

15. Wrote second shell script to automate the build and deployment process in Jenkins. it builds the WAR file using Maven, upload the timestamped build to an S3 bucket, and download the latest WAR file to the Tomcat webapps directory for deployment as LoginWebApp.war.

Command

See the list of available environment variables

```
#!/bin/bash

DATE=$(TZ="Asia/Kolkata" date +%d-%m-%Y)
TIME=$(TZ="Asia/Kolkata" date +%H-%M)
BUCKET_NAME="vj-assignment-5"

WAR_NAME="LoginWebApp_${DATE}_${TIME}.war"

cp /mnt/project/target/LoginWebApp.war /mnt/project/target/$WAR_NAME
aws s3 cp /mnt/project/target/$WAR_NAME s3://$BUCKET_NAME/
```

16. Wrote third script which removes the existing LoginWebApp files from the Tomcat webapps directory, retrieves the latest WAR file from the specified S3 bucket, and deploys it to Tomcat with the standard name LoginWebApp.war. This ensures that the most recent build is automatically deployed.

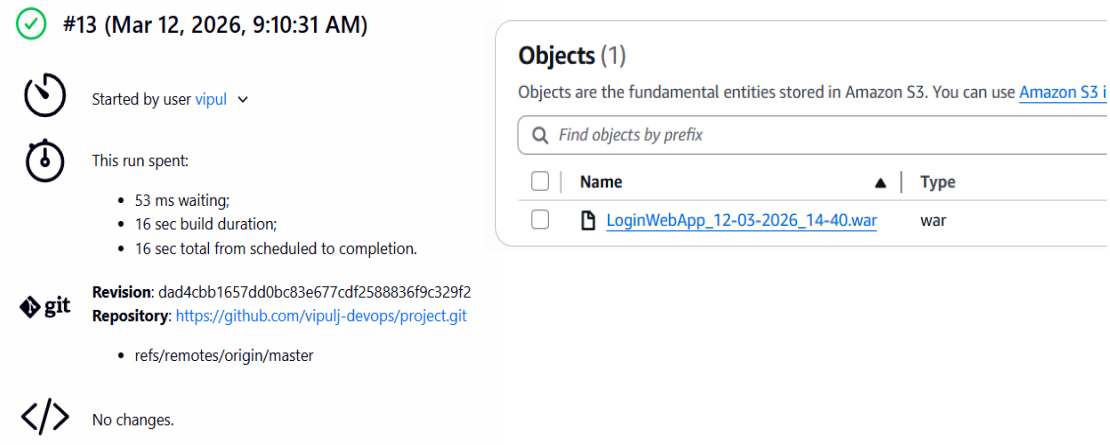
Execute shell ?

Command

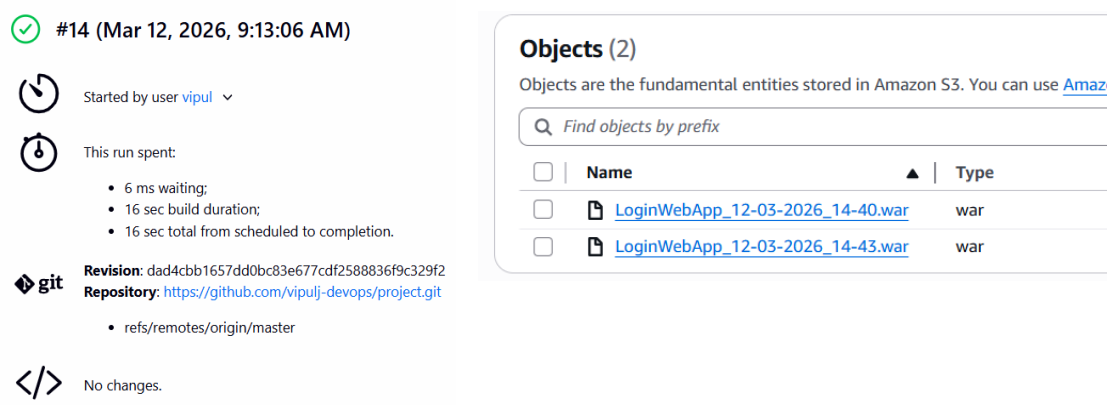
See the list of available environment variables

```
#!/bin/bash
BUCKET_NAME="vj-assignment-5"
rm -rf /mnt/servers/apache-tomcat-10.1.52/webapps/LoginWebApp*

LATEST_FILE=$(aws s3 ls s3://$BUCKET_NAME/ | sort | tail -n 1 | awk '{print $4}')

aws s3 cp s3://$BUCKET_NAME/$LATEST_FILE /mnt/servers/apache-tomcat-10.1.52/webapps/LoginWebApp.war
```

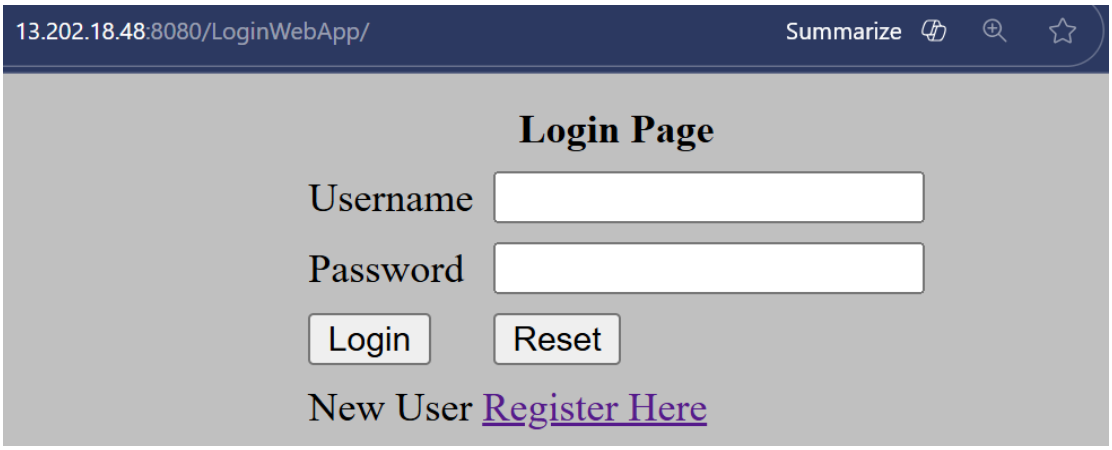
17. After that I applied the changes and saved the build configuration and then clicked on build now option. I did so many trial and error and on 13th trial the build ran successfully with no error. In the below image you can see after the build ran successfully the LoginWebApp war file got uploaded with timestamped name in S3 bucket.



18. On 14th build you can see the second war file also got added in the bucket with updated timestamp.



19. the latest WAR file is downloaded from the S3 bucket to the Apache Tomcat webapps directory and deployed with the standard name LoginWebApp.war.

```
[root@ip-172-31-33-28 ec2-user]# cd /mnt/servers/apache-tomcat-10.1.52/webapps/
[root@ip-172-31-33-28 webapps]#
[root@ip-172-31-33-28 webapps]# ls
LoginWebApp  LoginWebApp.war  ROOT  docs  examples  host-manager  jenkins  jenkins.war  manager
```

20. The latest war file got deployed on tomcat server.